

**USE OF WRENCH TO UNSCREW DRILLING RODS OR THE HAMMER
WITH A COME ALONG**

OBCTIVES

The purpose of this procedure is to define safe methods for using wrench to unscrew stuck rods

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

The staff with authority for the interpretation of this document is the Director of Operations. Any request for modification, revision or update must be made to one of this individual.

RESPONSABILITIES

The worker responsible of drilling must ensure that this procedure is followed and inspect the necessary tools and coordinate the task.



**NECESSARY
TOOLS**

- 1 come along
- N.B. The hooks must be equipped with latches.
- 2 shackles 3/4
- Rod wrenchs
- Saver sub wrench
- Sledgehammer

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STEPS

1. Place the flat notches in the upper rod then opposite the boom anchor and the bottom rod in the 'flat slip'
2. Apply the lock out tag out procedure;
3. Insert the wrench into the notches in the rods.
4. Install one of the come along hooks on the boom anchor and the other on the wrench shackle.
5. Start tightening the come along to put tension on the jammed drill rod.
6. Tighten the come along until the rods separate.
N.B: It may be necessary to hit the handle of the wrench with a sledgehammer to separate the rods.

In this case, make sure your feet are on a solid surface and that no one is in line if the hammer is dropped.



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This operation can also be performed without using the boom anchor. The same principle can be used by taking two wrenches. In this case simply connect the come along to each end of the wrench and put a tension.

Unscrewing a rod or the hammer jammed in the saver sub

1. Put the **boom in horizontal position** to prevent the saver sub wrench from falling;
2. Place the gearbox as far as possible from tables to have enough space to install the wrench.



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3. Apply the lock out tag out procedure;
4. Install the saver sub wrench on it by passing it through the hammer or the rod;

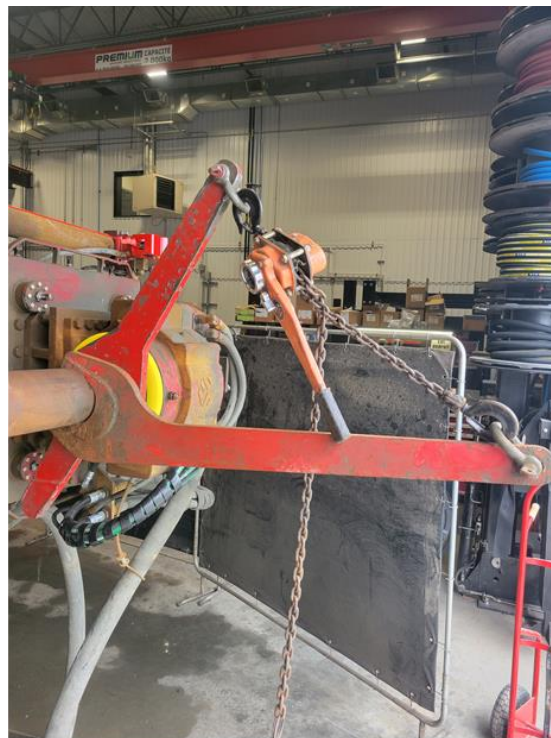


5. Install the rod wrench on the rod or on the hammer;



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6. Install one of the hooks of the come along on each of the shackles of the handles of the wrench;



7. Operate the come along until the rods separate.
N.B: It may be necessary to hit the handle of the wrench with a sledgehammer to help the rods to unscrew;
8. Remove the two wrenches and unlock the equipment then continue to work.

PROCEDURE FOR UNSCREWING THE HAMMER FROM THE DRILL STRING AT THE BOTTOM OF THE STOPE (WHEN BREAKING THE HOLE)

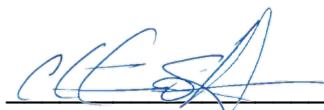
Although we favor unscrewing the 6" hammer at the level where the drill rig is located, some situations will require the hammer to be unscrewed from the drill string at the bottom of the stope. When this is necessary, a risk analysis must be carried out, and the following points must be specifically analyzed:

- Lockout of drill rig
- Gravitational energy of rods
- Communication between the 2 workers assigned to the task
- Condition of the back at the bottom of the stope
- Condition of the floor at the bottom of the stope
- Tools required for the job

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- Hammer line of fire after being unscrewed
- Space required for work
- Retrieving, handling and transporting the unscrewed hammer

Adds and/or modifications	Date	By
Addition of the sequence for rod or hammer jammed in the saver sub	July 2022	S. Tremblay
Changing "KEYS" for wrench Change some sentences to clarify	December 2023	S. Tremblay
Add lock out tag out at point #2	October 2024	S. Tremblay
Add risk analysis "and the hammer" in the title	March 2025	S. Tremblay Operations



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